

Vol 4 Chapter 10 — Dark Energy and Dark Matter

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Legacy chapter (May 2026). Last accuracy-checked 2026-09-11 (Keeper, K1892): the presentation scan found no retired reading here, and the tiers in the front matter are the May tiers, not re-verified line by line. Where this chapter and the Spine (Curriculum/Spine_DIV5_QM_GR_SM/) disagree, the Spine is current; where the Spine and the theorem registry disagree, the registry wins.

Dark energy (responsible for accelerated cosmic expansion) and dark matter (responsible for galaxy rotation curves and cosmic structure formation) together account for about 95% of the universe's energy density. Standard physics has no derivation of either; both are empirical inputs in the Λ CDM model.

BST derives both from substrate structure.

10.1 Dark energy as substrate vacuum

Dark energy in BST is the cosmological constant Λ of Chapter 4 — the substrate's Zone-2 Bergman vacuum integrated at cosmological scale. No additional dynamical dark-energy field is required; the substrate's vacuum is what produces accelerated expansion.

10.2 Dark matter as substrate Shannon channel

Dark matter has no substrate-side particle counterpart. The Five-Absence Prediction Set (Volume 2 Chapter 11) rules out a sterile DM particle. Instead, dark matter signatures are substrate Shannon channels — substrate-internal information channels carrying gravitational coupling without producing electromagnetic interactions. Galaxy rotation curves and cosmic structure formation emerge from the substrate's Shannon-channel gravitational coupling, with no need for unseen particles.

10.3 The dark-matter-to-baryon ratio

$$\frac{\Omega_{DM}}{\Omega_b} = \frac{3n_C + 1}{N_c} = \frac{16}{3} \approx 5.33.$$

Measured ratio (Planck): $\Omega_{DM}/\Omega_b \approx 5.36$. BST prediction matches at 0.58%. The structural reading uses $n_C = 5$ and $N_c = 3$ via the integer combination $(3n_C + 1)/N_c$ that emerges from the substrate's Shannon-channel allocation at the cosmological scale.

10.4 No WIMP, no axion, no sterile neutrino

The framework's prediction is that no dark matter particle exists. Experimental searches for WIMPs (direct-detection experiments LUX, XENON1T, etc.) and axions and sterile neutrinos will continue to find nothing. This is one of BST's sharpest beyond-Standard-Model predictions.

10.5 What comes next

Chapter 11 — gravitational waves and NANOGrav.

Where to look this up: DM/baryon ratio: substrate Shannon channel framework + integer derivation. For dark-matter experimental status: Particle Data Group annual reviews. For dark-energy reviews: Frieman, Turner, Huterer 2008, *Annual Review of Astronomy and Astrophysics* 46:385.